

STUDENT FEEDBACK MANAGEMENT SYSTEM

web-based application for collecting, storing, searching, filtering, and analyzing student feedback using HTML, CSS, Node.js, Express.js, and MongoDB.

Web Technology with Laboratory – Case Study 1

Course Code: 24ITPW501

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Program: B.Tech Information Technology

1. Introduction

The Student Feedback Management System is a web-based application developed to collect and manage student feedback about courses and faculty members. Students submit their details, course information, faculty name, rating, and comments through an online form. The submitted information is stored in MongoDB and can be managed through an administrator dashboard.

2. Objectives

- Collect student feedback through an online form.
- Store feedback records in MongoDB.
- Display all submitted feedback to the administrator.
- Search feedback using Register Number.
- Filter feedback based on Course.
- Calculate the average rating for a faculty member.
- Delete unwanted feedback records.
- Provide a simple and user-friendly interface.

3. Technologies Used

Technology	Purpose
HTML	Structure of student and admin pages
CSS	Styling and responsive layout
JavaScript	Client-side interaction and API calls
Node.js	Backend runtime environment
Express.js	Server and REST-style routes
MongoDB	Database for storing feedback
Mongoose	MongoDB schema and database operations

4. System Modules

Module	Description
Student Feedback	Students enter personal, course, faculty, rating, and comment details.
Feedback Storage	Express.js receives the form data and Mongoose stores it in MongoDB.
Admin Dashboard	Administrator views all feedback records in a table.
Search and Filter	Feedback can be searched by Register Number and filtered by Course.
Faculty Analysis	Average rating is calculated for a selected faculty member.
Delete	Administrator can remove a feedback record.

5. Database Design

Database: StudentFeedbackDB

Collection: feedback

Field	Type	Description
studentName	String	Student name
registerNo	String	Student register number
department	String	Department name
course	String	Course name
faculty	String	Faculty name
rating	Number	Rating from 1 to 5
comments	String	Student feedback comments

6. Application Workflow

- Student opens the feedback form at the local web server.
- Student enters the required details and submits the form.
- Express.js receives the POST request at /feedback.
- Mongoose saves the feedback document in the MongoDB feedback collection.
- The administrator opens the admin page to view the stored records.
- The administrator can search, filter, calculate faculty averages, and delete records.

7. Screenshots

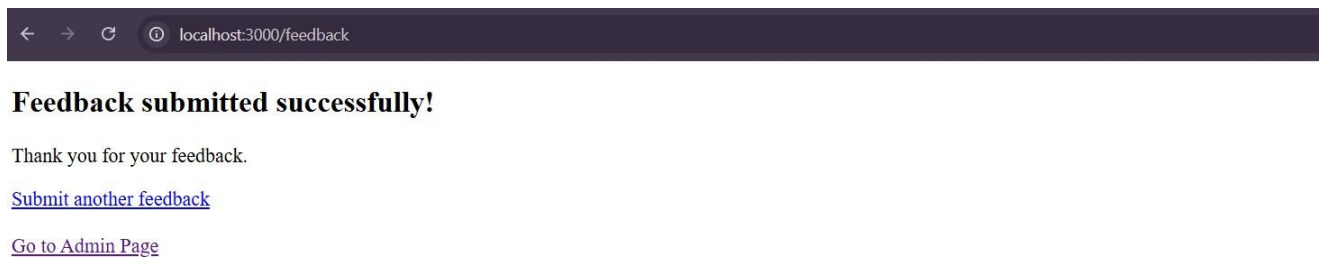


Figure 1: Student Feedback Form

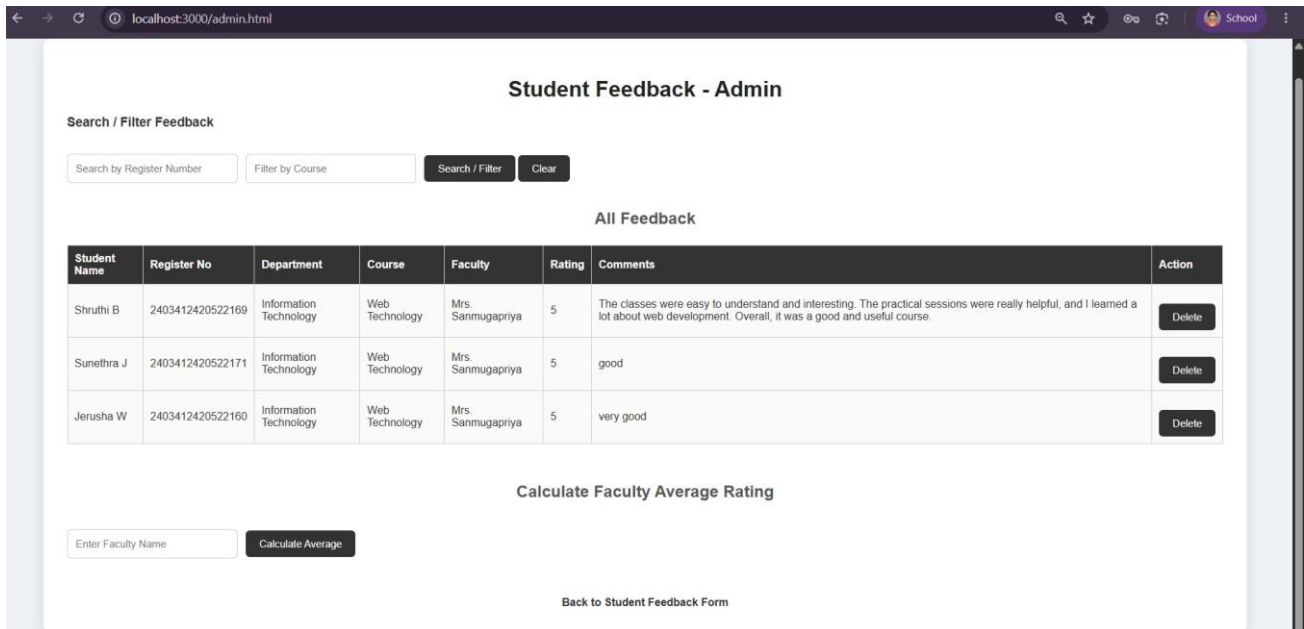


Figure 2: Feedback Submitted Successfully

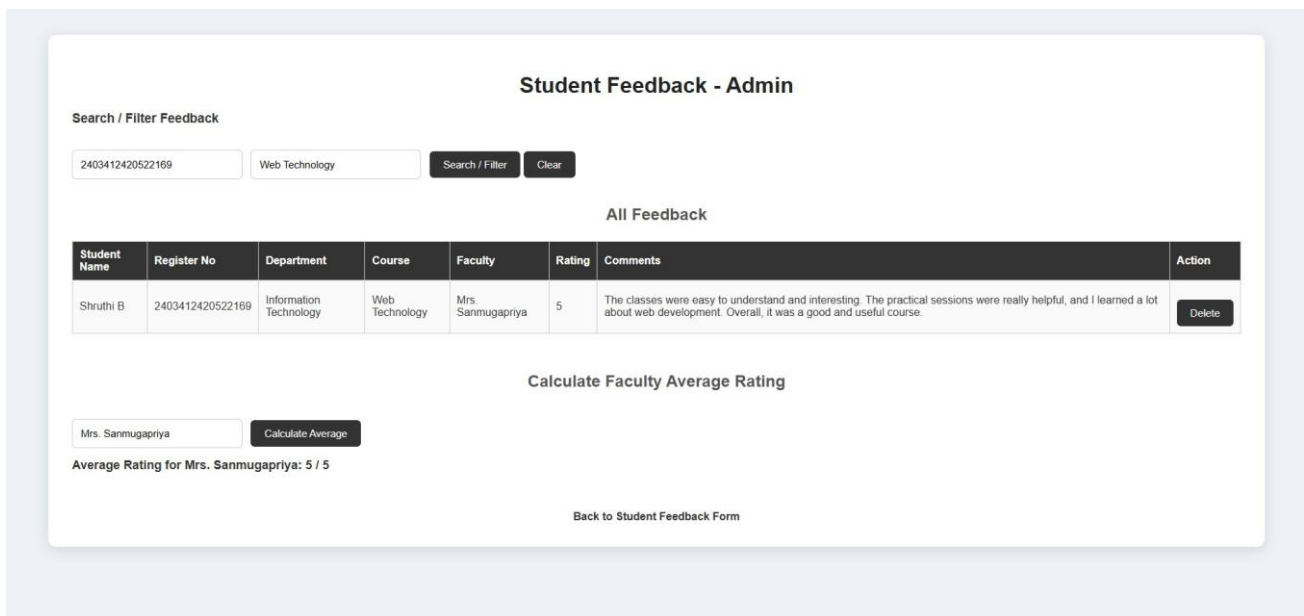


Figure 3: Admin Dashboard with Feedback Records

Student Feedback Management System

Student Name:

Register Number:

Department:

Course Name:

Faculty Name:

Rating:

Select Rating

Comments:

Submit Feedback

Figure 4: Admin Dashboard – Search/Filter and Faculty Average

8. Testing and Results

Test Case	Expected Result	Status
Submit feedback	Feedback saved successfully	PASS
View admin page	Stored records displayed	PASS
Search by Register Number	Matching record displayed	PASS
Filter by Course	Matching course records displayed	PASS
Faculty average	Average rating calculated	PASS
Delete feedback	Selected record removed	PASS

9. Result

The Student Feedback Management System was successfully implemented using HTML, CSS, JavaScript, Node.js, Express.js, Mongoose, and MongoDB. The application successfully accepts feedback, stores it in the database, displays records to the administrator, supports searching and filtering, calculates faculty average ratings, and allows deletion of records.

10. Conclusion

The project demonstrates how frontend forms can be integrated with a Node.js and Express.js backend and a MongoDB database. It provides a simple digital solution for collecting and managing student feedback and demonstrates important Web Technology concepts such as forms, routing, database operations, search, filtering, and dynamic data display.